

Scalability & Future Plan

Details	Information
Date	29/09/2026
Team ID	6ab4cbbb0fc666a751b50cda
Project Name	AI-Augmented Backend Development AI FitTrack API

Maximum Marks: 1 Mark

Scalability & Future Plan

AI FitTrack has a modular backend structure. In the future, the system can be improved to support more users, more workout data and additional fitness features.

Current System Limitations

S.No	Limitation	Impact	Priority
1	No deployment details are included in the current documentation.	The project is mainly described as a development/testing system.	Medium
2	Advanced analytics are not included.	Long-term progress tracking is limited.	Medium
3	Wearable integration is not included.	Activity data cannot be collected automatically.	Low

Scalability Plan

S.No	Aspect	Current State	Future Improvement
1	User Load	Node.js + Express API	Use load balancing and scalable hosting.
2	Data Storage	MongoDB + Mongoose	Use indexes, backups and managed database scaling.
3	Performance	Modular REST APIs	Add caching, monitoring and performance testing.
4	Security	JWT and bcrypt	Add rate limiting, better secret management and security monitoring.

Future Roadmap

Phase	Feature / Enhancement	Expected Benefit
Phase 2	Nutrition and BMI	More complete fitness tracking.
Phase 3	Wearable Integration	Automatic activity data.
Phase 4	Reminders and Analytics	Better progress monitoring and user engagement.